EXP NO: 1.A ANALYSE THE TREND OF DATA SCIENCE JOB POSTINGS OVER THE LAST DECADE

AIM:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

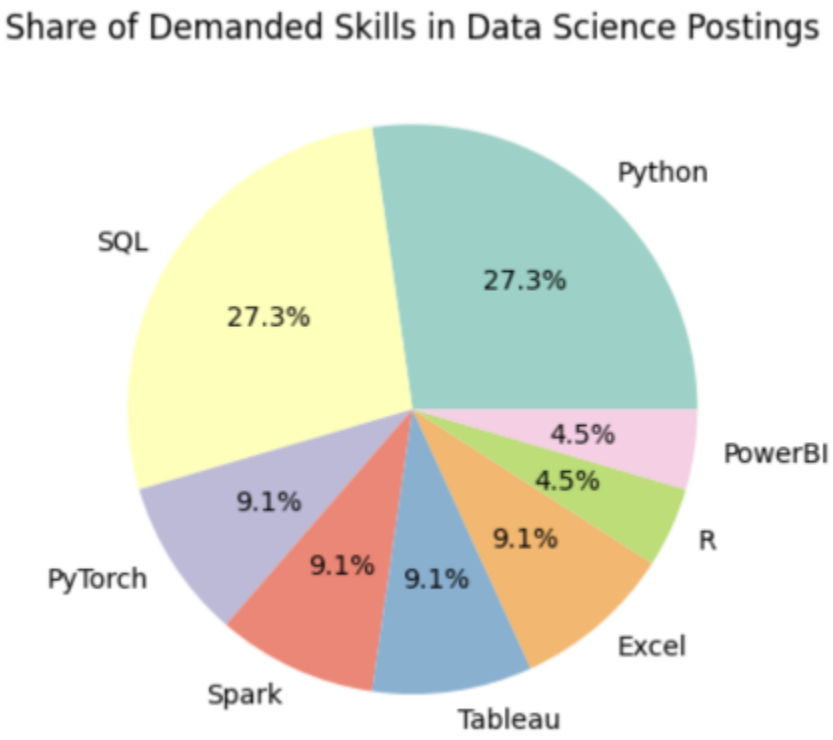
data = {
    'Job_Title': [
        'Data Scientist', 'Data Engineer', 'Data Analyst',
        'ML Engineer', 'Data Scientist', 'Data Analyst',
        'Data Engineer', 'Data Scientist'
    ],
    'Skills': [
        'Python, SQL, PyTorch',
        'Python, SQL, Spark',
        'SQL, Tableau, Excel',
        'Python, PyTorch',
        'Python, R, SQL',
        'Excel, SQL, PowerBI',
        'Python, Spark',
        'Python, SQL, Tableau'
    ]
}

df = pd.DataFrame(data)
skills_series = df['Skills'].str.split(', ').explode()
skill_counts = skills_series.value_counts()

plt.pie(
    skill_counts.values,
    labels=skill_counts.index,
    autopct='%1.1f%%',
    colors=plt.cm.Set3.colors
)
```

```
plt.title('Share of Demanded Skills in Data Science Postings')
plt.show()
```

OUTPUT:



RESULT:

Thus the code is executed successfully.

EXP NO: 1.B ANALYSE AND VISUALIZE THE DISTRIBUTION OF VARIOUS DATA SCIENCE ROLES FROM A DATA SET

AIM:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots

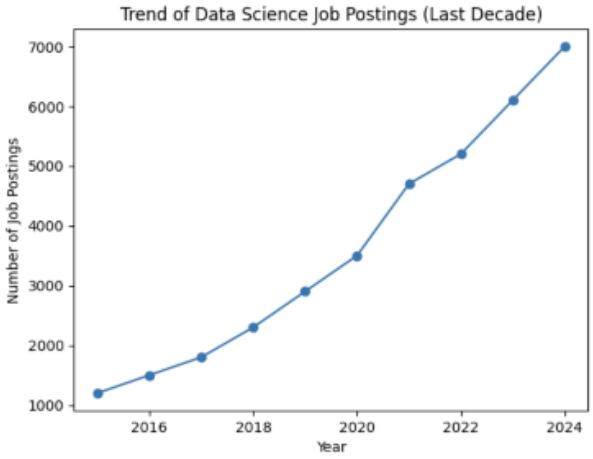
PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

data = {
    'Year': [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024],
    'Job_Postings': [1200, 1500, 1800, 2300, 2900, 3500, 4700, 5200, 6100, 7000]
}

df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job_Postings'], marker='o')
plt.title('Trend of Data Science Job Postings (Last Decade)')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

OUTPUT:



RESULT:

Thus the code is executed successfully.

EXP NO: 1.C CONDUCT AN EXPERIMENT TO DIFFERENTIATE STRUCTURED, UN-STRUCTURED AND SEMI STRUCTURED DATA BASED ON DATA SET GIVEN

AIM:

Create small datasets for each type and explain their characteristics.

PYTHON CODE:

```
import pandas as pd

structured = pd.DataFrame({
   "Name": ["Alice", "Bob"],
   "Age": [20, 21]
})
print(structured)


semi = {
   "Name": "Alice",
   "Age": 20
}
print(semi)

text = "Artificial Intelligence is changing the world."
print(text)
```

OUTPUT:

```
Name  Age
0  Alice   20
1    Bob   21
{'Name': 'Alice', 'Age': 20}
Artificial Intelligence is changing the world.
```

RESULT:

Thus the code is executed successfully.

EXP NO: 1.D CONDUCT AN EXPERIMENT TO ENCRYPT AND DECRYPT GIVEN SENSITIVE DATA

AIM:

Use the cryptography library to encrypt and decrypt a piece of data.

PYTHON CODE:

```
from cryptography.fernet import Fernet

key = Fernet.generate_key()
f = Fernet(key)

data = "PIN-8#290"
print("Original Data:", data)

encrypted = f.encrypt(data.encode())
print("\nEncrypted Data:", encrypted)

decrypted = f.decrypt(encrypted).decode()
print("\nDecrypted Data:", decrypted)
```

OUTPUT:

```
Original Data: My ATM PIN is 1234

Encrypted Data: b'gAAAAABqchkp4FK4b5PjhhF2ZIW6jbLJ7zET9w3-
U57owpV_ICEw63YvtkpO5yTotmcgb5imtJCSlwyDQLmXXi8ABkVvkx5xJfmL
dVcfgBReBxPNikJbxlA='
```

RESULT:

Thus the code is executed successfully.